Maven practicle wIth Tomcat
Now we will do Maven + Tomcat.
So what is tomcat?
Apache Tomcat is a **Java-based web server and servlet container** used to run **Java web applications**.

Maven command to start project.
Step 1. Make a new directory <folder name> using mkdir <folder name>
Then cd <folder name>
Step 2. Write command:
mvn archetype: generate -DgroupId=com.demo -DartifactId=web-demo -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false

Now Explanation:
mvn archetype:generate → archetype = template generator
generate = create a new project

-DgroupId = your organization's name
-DartifactId = your project's name
-DarchetypeArtifactId → Create a **web application project**
-Dinteractive= false → now don't ask any further question just make project with the given instructions.

This will generate our Project with starting POM.xml

```
sumitkumarmehta@sumits-MacBook-Air lemaven % mvn archetype:generate -DgroupId=com.demo -DartifactId=web-demo -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 74 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 96 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml (1.1 kB at 41 kB/s)
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 92 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.0/maven-archetype-webapp-1.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.0/maven-archetype-webapp-1.0.jar (3.9 kB at 158 kB/s)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /Users/sumitkumarmehta/lemaven
[INFO] Parameter: package, Value: com.demo
[INFO] Parameter: groupId, Value: com.demo
[INFO] Parameter: artifactId, Value: web-demo
[INFO] Parameter: packageName, Value: com.demo
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /Users/sumitkumarmehta/lemaven/web-demo
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  0.934 s
[INFO] Finished at: 2026-03-31T12:17:45+05:30
[INFO] ----------------------------------------------------------------------------
sumitkumarmehta@sumits-MacBook-Air lemaven %
```

```
sumitkumarmehta@sumits-MacBook-Air lemaven % ls
dockermaven     web-demo
sumitkumarmehta@sumits-MacBook-Air lemaven %
```

```
sumitkumarmehta@sumits-MacBook-Air web-demo % mvn tomcat7:run
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/maven-metadata.xml (42
7 B at 1.6 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/
2.2/tomcat7-maven-plugin-2.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/2
.2/tomcat7-maven-plugin-2.2.pom (12 kB at 377 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat-maven-plugin/2
.2/tomcat-maven-plugin-2.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat-maven-plugin/2.
2/tomcat-maven-plugin-2.2.pom (33 kB at 1.2 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/12/apache-12.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/12/apache-12.pom (16 kB at 5
75 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/
2.2/tomcat7-maven-plugin-2.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/2
.2/tomcat7-maven-plugin-2.2.jar (114 kB at 2.7 MB/s)
[INFO]
[INFO] ------------------------< com.demo:web-demo >-------------------------
[INFO] Building web-demo Maven Webapp 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] >>> tomcat7:2.2:run (default-cli) > process-classes @ web-demo >>>
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ web-demo ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform depend
ent!
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ web-demo ---
[INFO] No sources to compile
[INFO]
[INFO] <<< tomcat7:2.2:run (default-cli) < process-classes @ web-demo <<<
[INFO]
[INFO]
[INFO] --- tomcat7:2.2:run (default-cli) @ web-demo ---
```

```
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-settings/2.2.1/maven-s
ttings-2.2.1.jar (49 kB at 57 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/jcl-over-slf4j/1.7.5/jcl-over-slf4j
1.7.5.jar (17 kB at 19 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/httpcomponents/httpclient/4.3.1/ht
pclient-4.3.1.jar (586 kB at 326 kB/s)
[INFO] Running war on http://localhost:8080/
[INFO] Creating Tomcat server configuration at /Users/sumitkumarmehta/lemaven/web-demo/target/tomcat
[INFO] create webapp with contextPath:
Mar 31, 2026 12:29:08 PM org.apache.coyote.AbstractProtocol init
INFO: Initializing ProtocolHandler ["http-bio-8080"]
Mar 31, 2026 12:29:08 PM org.apache.catalina.core.StandardService startInternal
INFO: Starting service Tomcat
Mar 31, 2026 12:29:08 PM org.apache.catalina.core.StandardEngine startInternal
INFO: Starting Servlet Engine: Apache Tomcat/7.0.47
Mar 31, 2026 12:29:09 PM org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler ["http-bio-8080"]
```

Step 3. Command mvn clean install “Delete old build files, then build the project fresh and store the final artifact locally.”

This cleans the old build from the target files and build new → 1. Compile java code
2. Test → runs unit test
3. Package
4. Install

```
sumitkumarmehta@sumits-MacBook-Air web-demo % mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< com.demo:web-demo >-------------------------
[INFO] Building web-demo Maven Webapp 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.5
.1/maven-war-plugin-3.5.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.5.
1/maven-war-plugin-3.5.1.pom (9.5 kB at 49 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.5
.1/maven-war-plugin-3.5.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.5.
1/maven-war-plugin-3.5.1.jar (83 kB at 2.0 MB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ web-demo ---
[INFO] Deleting /Users/sumitkumarmehta/lemaven/web-demo/target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ web-demo ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform depend
ent!
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ web-demo ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ web-demo ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform depend
ent!
[INFO] skip non existing resourceDirectory /Users/sumitkumarmehta/lemaven/web-demo/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ web-demo ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ web-demo ---
[INFO] No tests to run.
[INFO]
[INFO] --- war:3.5.1:war (default-war) @ web-demo ---
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.29
/plexus-interpolation-1.29.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.29/
plexus-interpolation-1.29.pom (4.7 kB at 148 kB/s)
```

```
[INFO] Packaging webapp
[INFO] Assembling webapp [web-demo] in [/Users/sumitkumarmehta/lemaven/web-demo/target/web-demo]
[INFO] Processing war project
[INFO] Copying webapp resources [/Users/sumitkumarmehta/lemaven/web-demo/src/main/webapp]
[INFO] Building war: /Users/sumitkumarmehta/lemaven/web-demo/target/web-demo.war
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ web-demo ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util
/1.9.22/maven-resolver-util-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/
1.9.22/maven-resolver-util-1.9.22.pom (2.2 kB at 86 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.
22/maven-resolver-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.2
2/maven-resolver-1.9.22.pom (23 kB at 810 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/
1.9.22/maven-resolver-api-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1
.9.22/maven-resolver-api-1.9.22.pom (2.2 kB at 85 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus
-utils-4.0.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-
utils-4.0.1.pom (7.8 kB at 313 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util
/1.9.22/maven-resolver-util-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/
1.9.22/maven-resolver-util-1.9.22.jar (196 kB at 5.0 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/
1.9.22/maven-resolver-api-1.9.22.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus
-utils-4.0.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1
.9.22/maven-resolver-api-1.9.22.jar (157 kB at 4.5 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-
utils-4.0.1.jar (193 kB at 1.8 MB/s)
[INFO] Installing /Users/sumitkumarmehta/lemaven/web-demo/pom.xml to /Users/sumitkumarmehta/.m2/repository/c
om/demo/web-demo/1.0-SNAPSHOT/web-demo-1.0-SNAPSHOT.pom
[INFO] Installing /Users/sumitkumarmehta/lemaven/web-demo/target/web-demo.war to /Users/sumitkumarmehta/.m2/
repository/com/demo/web-demo/1.0-SNAPSHOT/web-demo-1.0-SNAPSHOT.war
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.279 s
[INFO] Finished at: 2026-03-31T12:24:45+05:30
[INFO] ------------------------------------------------------------------------
```

Now You will see a “Build Success” Message

Step 4. Go to the localhost:8080 on any browser to see the webpage in action.

```
sumitkumarmehta@sumits-MacBook-Air web-demo % mvn tomcat7:run
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/maven-metadata.xml (42
7 B at 1.6 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/
2.2/tomcat7-maven-plugin-2.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/2
.2/tomcat7-maven-plugin-2.2.pom (12 kB at 377 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat-maven-plugin/2
.2/tomcat-maven-plugin-2.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat-maven-plugin/2.
2/tomcat-maven-plugin-2.2.pom (33 kB at 1.2 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/12/apache-12.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/12/apache-12.pom (16 kB at 5
75 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/
2.2/tomcat7-maven-plugin-2.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/2
.2/tomcat7-maven-plugin-2.2.jar (114 kB at 2.7 MB/s)
[INFO]
[INFO] ------------------------< com.demo:web-demo >-------------------------
[INFO] Building web-demo Maven Webapp 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] >>> tomcat7:2.2:run (default-cli) > process-classes @ web-demo >>>
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ web-demo ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform depend
ent!
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ web-demo ---
[INFO] No sources to compile
[INFO]
[INFO] <<< tomcat7:2.2:run (default-cli) < process-classes @ web-demo <<<
[INFO]
[INFO]
[INFO] --- tomcat7:2.2:run (default-cli) @ web-demo ---
```